University of 8 May 1945 - Guelma
Department of Computer Science

Solution to the Exam on Business Intelligence

Level: 4th Year AI - ING **Date:** 11/01/2026 **Duration:** 2h

Score	
..... / 20	**Full Name:** ..... **Group:** .....

Exercise 1: Multiple Choice Questions (14 pts)

Grading: Correct answer = **+0.4 pts**. Incorrect answer = **-0.15 pts**.
Unanswered = **0 pts**.

0.4 pts × 35 questions = 14 pts

(The total score is rounded up to the nearest 0.25)

Part I: BI Fundamentals & Concepts

Q1. What is the primary goal of Business Intelligence (BI) systems?

- ☐ **A**. To process daily operational transactions with high speed and ensure data integrity.
- ☒ **B**. To transform raw data into meaningful information and knowledge for decision-making.
- ☐ **C**. To replace human employees with automated Artificial Intelligence algorithms.
- ☐ **D**. To strictly store data for legal compliance without any analytical purpose.

Q2. How does a Data Mart typically differ from a Data Warehouse?

- ☐ **A**. A Data Mart is always larger in storage size than a DW.
- ☒ **B**. A Data Mart focuses on a specific department, whereas DW covers the entire enterprise.
- ☐ **C**. A Data Mart contains raw operational data, while DW contains summarized data.
- ☐ **D**. There is no difference; the terms are synonyms.

Q3. To ensure the success of a BI project, KPIs should follow the "SMART" approach. What does this acronym stand for?

- ☐ **A**. Strategic, Measurable, Automated, Reported, Time-based.
- ☐ **B**. Specific, Measurable, Attainable, Reliable, Time-based.
- ☒ **C**. Specific, Measurable, Attainable, Relevant, Time-based.
- ☐ **D**. Simple, Managed, Accurate, Relevant, Technical.

Q4. Which statement accurately describes the main difference between OLTP and OLAP systems?

- ☐ **A**. OLTP handles historical data, while OLAP handles current real-time data.
- ☐ **B**. OLTP is designed for strategic analysis, while OLAP is for data entry clerks[1].
- ☐ **C**. OLTP is read-intensive (handles complex queries), while OLAP is write-intensive (handles fast inserts).
- ☒ **D**. OLTP supports day-to-day operations, while OLAP supports analysis.

Q5. According to the definition, Decision Support Systems (DSS) are specifically designed to help solve which type of problems?

- ☐ **A**. Fully structured, routine operational tasks.
- ☒ **B**. Semi-structured problems relying on human judgment.
- ☐ **C**. Completely unstructured problems with no data.
- ☐ **D**. Automated tasks requiring no human intervention.

Part II: The ETL Process & Data Preparation

Q6. What is the primary purpose of the "Transformation" step in an ETL process?

- ☐ **A**. To load raw data directly into the data warehouse for backup purposes.
- ☐ **B**. To extract data from various sources like APIs and web pages.
- ☒ **C**. To process data so it fits the needs of the target system and the intended use case.
- ☐ **D**. To visualize the data using dashboards and reporting tools.

Q7. Which of the following scenarios best describes a "Push method" in data loading?

- ☐ **A**. The data warehouse subscribes to a source and requests data at scheduled intervals.
- ☒ **B**. The source system sends data to the warehouse as soon as an event occurs.
- ☐ **C**. A Python script queries a database every night to fetch new records.
- ☐ **D**. The ETL tool polls the API every hour to check for updates.

Q8. When is "Incremental Loading" preferred over "Full Loading"?

- ☐ **A**. When you are setting up a new data warehouse for the first time.
- ☐ **B**. When the dataset is small and changes completely every day.
- ☒ **C**. When you need to load only data that has changed since the last load.
- ☐ **D**. When the target system does not support data appending.

Q9. Which loading strategy would you use to run a stock update every day at midnight?

- ☐ **A**. Stream Loading via Push method.
- ☐ **B**. Real-time Incremental Loading.
- ☒ **C**. Batch Loading using Cron job.
- ☐ **D**. Full Loading manually triggered.

Q10. What does "Binning" refer to in Data Transformation?

- ☐ **A**. Converting text data into binary format (0s and 1s).
- ☒ **B**. Grouping continuous data into specific categories or intervals.
- ☐ **C**. Combining multiple values into a single summary value like sum or average.

[1] Operational staff who manually input daily data (e.g., cashiers, receptionists).

☐ **D**. Deleting rows that contain missing values.

Q11. Why is "Lossy Data Compression" (e.g., converting floating-point values to integers) considered a risk for information loss?

☐ **A**. It filters out rows, permanently discarding specific data.
☐ **B**. It aggregates multiple records, hiding individual details.
☒ **C**. It reduces precision, making original values unrecoverable.
☐ **D**. It encrypts data, making it unreadable without a key.

Q12. You have a DataFrame `df` with a column 'Age' containing some NaN values. Which Pandas command would you use to replace these missing values with the average age?

☐ **A**. `df['Age'].dropna(mean=True)`
☒ **B**. `df['Age'].fillna(df['Age'].mean(), inplace=True)`
☐ **C**. `df['Age'].replace(np.nan, 'Average')`
☐ **D**. `df['Age'].transform('mean')`

Q13. In an ETL pipeline, you need to convert a column 'Date_String' (e.g., "2023-01-01") into a proper datetime object. Which function should be used?

☐ **A**. `df['Date_String'].astype('date')`
☒ **B**. `pd.to_datetime(df['Date_String'])`
☐ **C**. `df['Date_String'].convert_datetime()`
☐ **D**. `pd.read_csv(df, parse_dates=True)`

Q14. You want to check the total number of missing values for each column in a DataFrame named `df`. Which command should you use?

☐ **A**. `df.isnull().count()`
☒ **B**. `df.isnull().sum()`
☐ **C**. `df.dropna().count()`
☐ **D**. `df.isna().value_counts()`

Q15. In the code `imputer.fit_transform(df[['order_value_EUR']])`, why are double brackets `[[...]]` used instead of single brackets? (Note: `imputer` is a `SimpleImputer` object).

☐ **A**. To select multiple columns from the dataset for simultaneous processing.
☐ **B**. To force the output to be a one-dimensional Pandas Series object.
☒ **C**. To pass a 2D array structure, which is required by Scikit-Learn.
☐ **D**. To fix a syntax error, since single brackets are invalid here.

Q16. What is the effect of executing `df = df.ffill()` on a column with missing values?

☐ **A**. It replaces missing values with the mean of the column.
☐ **B**. It replaces missing values with the value from the next row.
☒ **C**. It replaces missing values with the value from the previous row.
☐ **D**. It drops the rows containing missing values.

Q17. Consider the code: `df['city'] = df['city'].str.strip()`. What is the result of this operation on the string " New York "?

☒ **A**. "New York". ☐ **B**. "NewYork". ☐ **C**. "new york". ☐ **D**. "New York ".

Note: *The* `strip()` *removes leading and trailing whitespace only, but preserves spaces between words.*

Q18. Which of the following is NOT considered a data extraction technique?

☐ **A**. CCD Sampling. ☒ **C**. Data Mining.

☐ **B**. Edge Computing. ☐ **D**. Database Querying.

Note: *Data Mining is an analysis phase (Pattern Discovery) that happens after data is stored. Edge Computing, however, is a valid extraction/collection technique used to gather data from IoT sensors.*

Q19. Which of the following is NOT an example of "Normalizing"?

☐ **A**. Converting a column with mixed "meters" and "feet" values into "meters" only.

☐ **B**. Converting all currency values in a dataset to a single currency (e.g., USD).

☒ **C**. Converting a date string "01/01/2024" into a standard Date Object format.

☐ **D**. Converting temperature readings from Fahrenheit to Celsius.

Part III: Data Warehouse Modeling & OLAP

Q20. Why might a designer choose a Snowflake Schema over a Star Schema despite the complexity?

☐ **A**. To significantly increase query speed for simple aggregations.

☒ **B**. To reduce the memory footprint[2] by eliminating data redundancy in dimension tables.

☐ **C**. To make the model easier to understand and manipulate via drag-and-drop for business users.

☐ **D**. To allow for unlimited dimensions as per Codd's rules.

Q21. What is the primary structural difference between a Star Schema and a Snowflake Schema?

☐ **A**. A Snowflake Schema has multiple fact tables, whereas a Star Schema has only one.

☐ **B**. A Star Schema uses normalized dimension tables, whereas a Snowflake Schema uses denormalized dimensions.

☒ **C**. A Star Schema uses denormalized dimension tables, whereas a Snowflake Schema normalizes dimensions into hierarchies.

☐ **D**. A Snowflake Schema is faster for queries than a Star Schema because it has fewer joins.

Q22. Which OLAP operation corresponds to "Summarizing data along a dimension" (e.g., converting daily sales to quarterly sales using SUM or AVG)?

☐ **A**. Dicing. ☐ **B**. Pivoting. ☒ **C**. Rolling Up. ☐ **D**. Drilling Down.

Q23. Which OLAP operation involves moving from "Total Monthly Sales" to "Daily Sales per Store"?

☐ **A**. Rolling Up. ☐ **B**. Pivoting. ☐ **C**. Slicing. ☒ **D**. Drilling Down.

Q24. You want to analyze sales data for only the year "2018" across all states and products. Which OLAP operation is this?

☐ **A**. Dicing. ☒ **B**. Slicing. ☐ **C**. Rolling Up. ☐ **D**. Drilling Down.

Q25. According to Codd's 12 Rules for OLAP, what does "Transparency" (Rule 2) imply?

[2]Memory footprint: The amount of storage space or RAM required to store the data.

- ☐ **A**. The system must allow users to see the underlying SQL code for every query.
- ☒ **B**. The user typically does not need to know where the data is stored.
- ☐ **C**. The system must disclose all calculation formulas to the end-user.
- ☐ **D**. The system allows multiple users to work simultaneously without contradiction.

Q26. Which of the following best describes the content of a "Fact Table" in a data warehouse?

- ☐ **A**. It stores descriptive attributes used for filtering and grouping.
- ☒ **B**. It stores numerical metrics and foreign keys connecting to dimensions.
- ☐ **C**. It stores numerical metrics and all descriptive text in one place.
- ☐ **D**. It stores hierarchical structures to optimize storage space.

Q27. In the context of the Snowflake Schema, what is the primary goal of "Normalizing" a dimension table?

- ☐ **A**. To merge hierarchical attributes into a single table to speed up reading queries.
- ☒ **B**. To split a dimension into multiple related tables to reduce data redundancy and save storage.
- ☐ **C**. To standardize the data format (e.g., converting all dates to DD/MM/YYYY) to ensure consistency.
- ☐ **D**. To denormalize the data structure so that all attributes are directly accessible without joins.

Q28. In In Kimball's 4-Step method, what does determining the "Granularity" specifically refer to?

- ☐ **A**. Determining the total volume of data that the data warehouse will store.
- ☐ **B**. Determining the number of dimensions needed for the analysis.
- ☒ **C**. Determining the specific level of detail for the facts.
- ☐ **D**. Determining the depth of the hierarchy within a dimension.

Part IV: Analytical SQL & Performance Indicators

Q29. What is the main difference between a Window Function and a standard Aggregate Function with GROUP BY?

- ☐ **A**. Window functions cannot perform calculations like SUM or AVG.
- ☐ **B**. Window functions collapse multiple rows into a single summary row, losing individual details.
- ☒ **C**. Window functions calculate values across a group of rows but keeps each row separate.
- ☐ **D**. Window functions are only used for text manipulation, not numerical calculation.

Q30. If you use `SUM(Sales) OVER(ORDER BY Date)`, what result will you generate?

- ☐ **A**. The total sum of sales for the entire table on every row.
- ☒ **B**. A running total of sales.
- ☐ **C**. The average sales per day.
- ☐ **D**. The maximum sales value found in the table.

Note: *The inclusion of* `ORDER BY` *inside* `OVER()` *forces a cumulative calculation (Running Total) row by row. Option A would be the result if we used* `SUM(Sales) OVER()` *without sorting.*

Q31. Which SQL clause is mandatory when using Ranking Functions like `RANK()` or `ROW_NUMBER()`?

- ☐ **A.** PARTITION BY.
- ☒ **C.** ORDER BY.
- ☐ **B.** GROUP BY.
- ☐ **D.** HAVING.

Q32. In the context of Analytical SQL, what is the precise definition of a "Window"?

- ☒ **A.** It is a logical set of rows related to the current row that participates in a calculation.
- ☐ **B.** It is a group of rows that are collapsed into a single summary row to reduce the result set size.
- ☐ **C.** It is a visual interface or dashboard used to display performance indicators.
- ☐ **D.** None of these.

Q33. In the function `LEAD(DailySales, 2) OVER(...)`, what does the second parameter (2) represent?

- ☐ **A.** The Default Value: It is the value returned if there is no subsequent row found.
- ☒ **B.** The Offset: It specifies how many rows forward to look relative to the current row.
- ☐ **C.** The Partition Count: It divides the dataset into 2 separate groups.
- ☐ **D.** The Lag: It tells the function to look 2 rows behind instead of ahead.

Q34. Which of the following is true regarding Indicators vs KPIs?

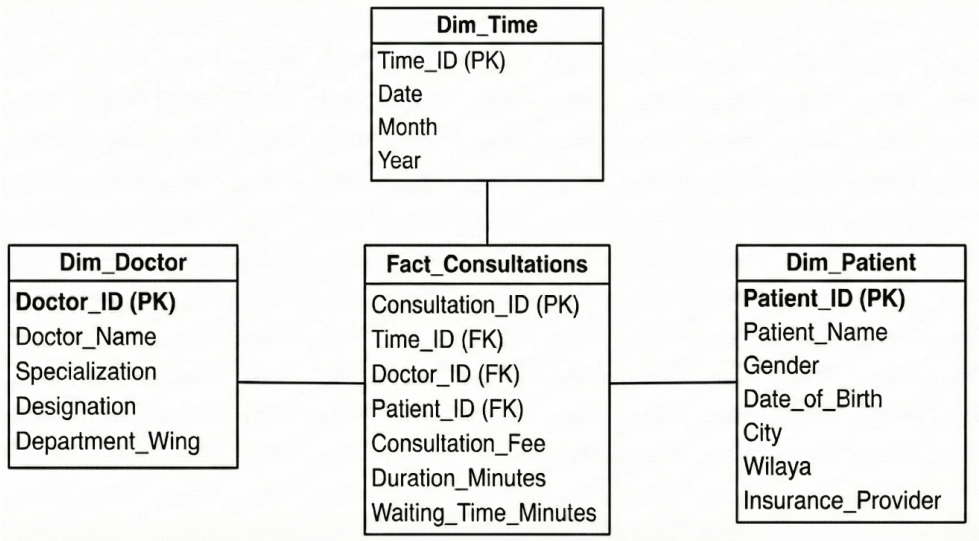
- ☐ **A.** All indicators are KPIs, but not all KPIs are indicators.
- ☐ **B.** Indicators are always tied to strategic goals, while KPIs are just operational.
- ☒ **C.** KPIs are specific measures that focus on what drives strategic success.
- ☐ **D.** An organization should have as many KPIs as possible (more than 20) to be safe.

Q35. Which of the following is NOT a key benefit of using Dashboards?

- ☐ **A.** Instant, at-a-glance visibility.
- ☐ **C.** Interactivity (slicing and drilling down).
- ☐ **B.** Rapid detection of issues and trends.
- ☒ **D.** Strictly static, read-only reporting.

Exercise 2: DW Modeling & Analytical SQL (6 pts)

Context: City Care Hospital's data warehouse tracks daily patient consultations. The schema consists of three dimensions: **Patient**, **Time** and **Doctor** (the latter is characterized by Specialization e.g., Cardiology, Pediatrics; Designation e.g., Senior Consultant, Resident; and Department Wing e.g., North Block). There are also three measures (facts): *Consultation fee*, *duration of the consultation* and *waiting time*. The complete structure including facts and attributes is illustrated in the Star Schema below:



1. Identify the **grain** of the `Fact_Consultations` table in the provided Star Schema. Explain the reasoning behind your determination. **(1 pt)**

   Answer: The grain is a single patient consultation (or one individual consultation event).

   Reasoning: The presence of the unique primary key `Consultation_ID` in the fact table indicates that each row represents a distinct transaction (a specific visit), rather than aggregated data. Furthermore, the measures (Fee, Duration, Waiting Time) apply to a single event.

   Marking Scheme:
   - **[0.5 pt] Identification:** Awarded for correctly stating "Consultation", "Patient Visit", or "Transaction".
   - **[0.5 pt] Justification:** Awarded for mentioning the Primary Key (`Consultation_ID`) or the non-aggregated nature of the data.

2. Is there a dimension table where we can apply **Normalization**? If yes, identify the dimension and specify the name and attributes of the new table that would be created (just give one example).**(1 pt)**

   Answer: Yes, we can apply Normalization. For Example, we can normalize the Dim_Patient table.

   - New Table Name: `Dim_City`
   - Attributes: `City_ID` (PK), `City_Name`, `Wilaya`.

   Marking Scheme:
   - **[0.25 pt] Identification:** Awarded for identifying a valid dimension to normalize (e.g., `Dim_Patient`, `Dim_Doctor`, or `Dim_Time`).
   - **[0.75 pt] Specification:** Awarded for correctly defining the new table structure (Must include a **Primary Key** and relevant attributes).
   - *Note: Mentioning the Foreign Key link in the original table = BONUS +0.25pts*
   - **Alternative Answers:** Other valid answers are accepted, such as: Normalizing `Dim_Doctor` by creating a `Dim_Specialization` table (containing *Specialization* and *Designation*); Normalizing `Dim_Patient` by creating a `Dim_Insurance` table (to store unique *Insurance_Provider* names); Normalizing `Dim_Time` by splitting *Year* and *Month* into a separate hierarchy (e.g., `Dim_Year`).

3. In `Dim_Time`, attributes like Month and Year are stored physically as columns. Why do we store them explicitly instead of just keeping the **full date**? **(0.75 pt)**

   Answer: We store them explicitly **to improve query performance**.

   Calculating the year or month from a full date at runtime (e.g., using YEAR(Date)) is computationally expensive for large datasets. Storing them as explicit physical columns makes filtering and grouping significantly faster.

   Marking Scheme:
   - **[0.25 pt] Keyword:** Awarded for mentioning "Performance", "Speed", "Optimization", or "Efficiency".
   - **[0.5 pt] Technical Reason:** Awarded for explaining the mechanism (e.g., "Avoids runtime calculation", "Enables Indexing", or "Faster GROUP BY/Filtering").

4. **Consider the following SQL query:**

```sql
SELECT * FROM (
    SELECT Doctor_Name, Consultation_Fee,
           RANK() OVER (ORDER BY Consultation_Fee DESC) as Ranking
    FROM Fact_Consultations
    JOIN Dim_Doctor ON Fact_Consultations.Doctor_ID = Dim_Doctor.Doctor_ID
    )
WHERE Ranking <= 3;
```

a. **What exactly does this query return?** (Describe the business result). **(0.5 pt)**

It returns the **top 3 most expensive consultations** (highest fees) recorded in the hospital, displaying the name of the doctor who performed each one.

Marking Scheme:

- **[0.5 pt] Interpretation:** Awarded for identifying "Highest Fees" or "Most Expensive Consultations".
- *Critical Note:* Answers stating **"Top 3 Doctors with highest total revenue"** are **INCORRECT**. The query ranks individual transactions, not aggregated doctor totals (There is no `GROUP BY`).

b. **Why is the subquery (nested SELECT) necessary here?** Why couldn't we simply use `WHERE Ranking <= 3` directly in the main query without nesting? **(0.75 pt)**

Answer: Because of the SQL Logical Order of Execution.

Reasoning: Window functions like `RANK()` are calculated during the `SELECT` phase, which happens **after** the `WHERE` clause is processed. Therefore, the alias "Ranking" does not exist yet when the `WHERE` clause runs. We must use a subquery to "materialize" the rank first so the outer query can filter it.

Marking Scheme:

- **[0.5 pt] Concept:** Awarded for mentioning "Order of Execution" or stating that "`WHERE` runs before `SELECT` (or `RANK`)".
- **[0.25 pt] Consequence:** Awarded for explaining that the alias "Ranking" is not yet available/recognized in the WHERE clause.

5. Using **Window Functions**, write the SQL queries to calculate the following KPIs:

a. **Service Speed Comparison:** Display the `Waiting_Time_Minutes` of each individual consultation alongside the **average Waiting Time** for that specific **Doctor** (for comparison). **(1 pt)**

Answer:

```sql
SELECT
    Dim_Doctor.Doctor_Name,
    Fact_Consultations.Waiting_Time_Minutes,
    AVG(Waiting_Time_Minutes) OVER (PARTITION BY Doctor_ID) as
   Avg_Wait_Per_Doc
FROM Fact_Consultations
JOIN Dim_Doctor ON Fact_Consultations.Doctor_ID = Dim_Doctor.Doctor_ID;
```

Marking Scheme:

- **[0.5 pt] Basic Query:** Awarded for correct SELECT columns and JOIN between Fact and Doctor.
- **[0.5 pt] Window Function:** Awarded for correctly using `AVG(...) OVER (PARTITION BY ...)`.

b. Contribution by Department: Calculate the revenue generated by each **Doctor** as a **percentage (%)** of the total revenue of their specific **Department Wing** (`Department_Wing`). **(1 pt)**

Answer:

```sql
WITH Doctor_Stats AS (
    SELECT D.Doctor_Name, D.Department_Wing,
           SUM(F.Consultation_Fee) as Doc_Revenue
    FROM Fact_Consultations F
    JOIN Dim_Doctor D ON F.Doctor_ID = D.Doctor_ID
    GROUP BY D.Doctor_Name, D.Department_Wing
)
SELECT
    Doctor_Name,
    Doc_Revenue,
    (Doc_Revenue / SUM(Doc_Revenue) OVER (PARTITION BY Department_Wing)
   ) * 100
           as Pct_Contribution
FROM Doctor_Stats;
```

Marking Scheme:

- **[0.5 pt] Aggregation:** Awarded for correctly calculating the individual Doctor's Revenue.
- **[0.5 pt] Window Function Implementation:**
  - **Full Mark (0.5):** Uses `OVER (PARTITION BY ...)` correctly.
  - **Partial Mark (0.25):** Uses a standard **JOIN** or **Subquery** to get the correct result but ignores the instruction to use Window Functions.
  - **Zero (0.0):** Incorrect logic or syntax.